Clovis SFEIR

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Profile

IMT Atlantique engineering student (AI engineer apprentice at Infotel's AI Lab) who builds simulation, sensor-analysis and ML systems in C++20, Python and Rust, and delivers them as tested prototypes with measured results. Built a finite-difference PDE solver (OpenMP, **2nd-order convergence** verified), an async sensor-fusion pipeline (**0/10 false positives**), an IEC-104 protocol analyser (**39.5 ns**/message), and a from-scratch LoRA training engine.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*

Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France ***Class rank: 1/85*** | *Software Engineering, Databases, Cybersecurity*

HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlAce (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Reservoir Diffusivity PDE Solver *C++20, OpenMP, finite-difference*

- Explicit FTCS and implicit Crank-Nicolson finite-difference schemes for the reservoir diffusivity (pressure-transient) equation; verified via grid-refinement convergence study against the closed-form erfc analytical solution – error ratio converges to **4.0** as dx halves (textbook 2nd-order accuracy) across 5 refinements.
- OpenMP-parallelized stencil/assembly loops; measured (not assumed) scaling identifies the Amdahl's-law-bounded sequential Thomas solve and memory-bandwidth-bound behavior below L3 size; 13 test cases, 40,300 assertions; Docker image runs the full suite at build time.

LoRA Fine-Tuning Engine *Python, PyTorch, from-scratch transformer*

- Hand-written decoder-only transformer and LoRA layer on raw PyTorch autograd (no PEFT/Trainer); LoRA math checked against hand-computed values, plus a test that frozen base weights stay non-trainable.
- Measured 3-way comparison (no fine-tune / full / LoRA r=8): LoRA trains only **3.85%** of parameters (24.6K vs 614.8K) and forgets less (generic val. loss 2.20 vs 3.08 for full fine-tuning) but adapts less to the new domain (2.33 vs 0.80); 13 tests.

IIoT Predictive Maintenance *Rust, tokio, anomaly detection (EWMA/CUSUM)*

- Async telemetry pipeline (`tokio` mpsc) fuses EWMA + CUSUM control charts across 3 sensor channels to detect developing bearing-wear signatures, with linear-trend remaining-useful-life estimation.
- Sensor-fusion tuning (2-of-3 channels + widened control limits) cut false positives from a 0.41–0.50 precision baseline to **0/10 across 10 seeds**; precision **1.0000**, recall 0.9642, F1 0.9818, 43-sample latency; **1.6 M samples/s**; 26 tests, 0 failures.

ICS Intrusion Detection (IEC 60870-5-104) *C++20, intrusion/anomaly detection, explainable rules*

- Rule-based detector for IEC 60870-5-104 (no auth/encryption by protocol design) – the actual telecontrol protocol behind the 2016 Kyiv blackout – mapped to real attack patterns: control commands to unexpected breaker objects (the exact method used), sequence-number gaps (replay/desync), u-format/command floods.
- Parses APCI+ASDU at **39.5 ns** (**25.3 M msg/s**); end-to-end parse+detect 337 ns; **100% recall, 0 false positives** on a labeled synthetic trace (100 records, 15 attacks across 5 rules); 153 assertions/37 tests, ASan+UBSan clean.

Work Experience

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests; validated end-to-end producing a fully working, 100%-test-passing application directly from a single specification.

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

Languages	C++20, Python, Rust, C, Java, Bash, TypeScript, SQL; also Haskell, C#/.NET, Ada
Perf / Systems	OpenMP, cache-aware layout, acquire/release atomics, ICS/SCADA protocol parsing (Modbus, IEC-104)
Numerics / ML	finite-difference PDE, Monte Carlo, Kalman filter, EWMA/CUSUM change-detection, PyTorch/LoRA, BM25 retrieval
Infrastructure	Linux, Git, Docker, tokio async, GitHub Actions CI, Prometheus/Grafana, GCP, Redis, ZeroMQ
Frameworks	FastAPI, Flask, Node.js/Express, React/Next.js, REST/gRPC